From the

Annals of the White Tower Society

For the Year of the Hourglass, C.R. 930



ON THE DRAWING OF POWER BY STRING-SPELLS.

By Westphal, of White Tower, Fellow of the White Tower Society.

White Tower: Printed for the Society
C.R. 930

VII.

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Read before the Society in the Month of Frostfall, C.R. 929; printed in the Annals for the Year of the Hourglass, C.R. 930.

It is the common opinion, which I shall call the doctrine of two layers, that the world consists of the elements and of Tiamat only, and that the lines of law are Tiamat itself; that elemental workings draw their power through the elements, and that law-workings and string-spells alike draw theirs directly from the lines of law. On this opinion a law-working and a string-spell draw from one and the same place, and whatever disturbs the lines of law must disturb them both.

Yet it has long been reported among casters that a place of warding against magic will often fail to hold a string-spell. The older explanation is that string-spells are strong and so break the ward. But a string-spell of a low ring is not necessarily stronger than a law-working of a high one; and in the same warded place the latter cannot be cast while the former is cast as easily as anywhere. This difference is not to be explained by strength. The doctrine of two layers requires the two workings to be disturbed together; the report shows them to behave quite otherwise in a warded place.

The friends of the doctrine have offered two other explanations of the report. Some say that a ward is never perfectly close, and that the string-spell draws its power through the gaps. If this were so, a string-spell should fail in some place more strictly warded; yet no one has ever found a ward that holds it. Others say that string-spells are worked in the tongue of dragons, which the lines of law cannot lock. This is merely to give the matter a new name. It does not explain why locking the lines should not lock the dragon tongue; and if the dragon tongue does indeed pass by the lines of law, it concedes that the string-spell draws its power from some other place.

There is a further circumstance. Where the Babel fragments speak of a web, they use three different words. Former interpreters took the three for one, and rendered each as Tiamat. But the same fragments say that Masha wove Tiamat upon the Sea of Mana. If Tiamat were no more than a layer of the lines of law, then what was woven upon the sea and what the law-mage plucks would be one thing, and the fragments would have had no occasion for three words.

These things led me to suppose that a law-working and a string-spell draw their power from different places; and further, that every working draws its power through several layers lying one upon another, and the higher it takes hold, the more layers its power must pass. I shall raise this supposition, which I call the doctrine of layers, to the rank of a premiss, and I shall show that the assumption that the lines of law are Tiamat is superfluous for the explanation of what is observed. The difficulty of the doctrine of two layers lies in a matter that has never been thought through: by what right one asserts that two workings draw their power from the same place. With that matter I begin.

The First Part (§§ 1–5) explains what is meant by disturbance, how the places whence power is drawn are to be discerned, how the order of the layers may be read from the manner of disturbance, and how losses compound when several layers are disturbed together. The Second Part (§§ 6–14) records the trials, answers the objections that may be raised, establishes the doctrine upon the trials, and draws some consequences.

PART I. — OF DISCERNMENT

§ 1. Of Disturbance

When a working is cast in some surrounding, and its expenditure or its steadiness differs from what it shows when cast in a place of comparison, and the difference appears again and again on repetition and exceeds the variation among the comparisons themselves, I say that the working is disturbed in that surrounding.

There is as yet no instrument that weighs the mana spent in a single casting, and the expenditure must therefore be measured indirectly. Charged crystals are taken from one vein, none of them used before; the caster casts the same working again and again, drawing upon the power of a single crystal only, until its light goes out, and the number of castings is recorded. The fewer the castings, the greater the expenditure. The light is reckoned out when the face of the crystal, held under the same shaded lamp, no longer shows its grain; a crystal once spent is not used again. Every casting is counted, whether it takes form or not and whether it deviates or not; but a last casting begun as the light goes out, which takes no form, is not counted.

That the caster draws upon the crystal alone is tested thus: before and after each day's castings he tries his own remaining strength with one and the same small working, and finds it undiminished. This trial can discern a loss of about two or three castings of C, and no smaller; nor can it discern what he may recover between castings. That all that was spent came from the crystal is therefore an approximation only; but comparison and disturbance are cast to the same timetable, without rest within a round and with an hour between rounds, so that the approximation is the same on both sides.

One point more. The crystal is only the vessel that supplies the power for the time being. I assume that a working cast upon a crystal's power still takes hold at its own layer and draws through the same path, the power stored in the crystal having come in the end from the Sea of Mana; were it otherwise, the trials would measure only how a crystal supplies different workings, and not the places whence they draw. To test this we asked the casters to cast the three workings in storm and within the array upon their own strength, some hundreds of times each, counting not the castings but the deviations only: A deviated more in both places, B within the array only, and C in neither, as when cast upon crystals. These hundreds of castings were spread over more than ten days, no more than twenty or thirty of each working in a day, lest the casters tire; the same castings in a calm place served for comparison.

Steadiness is measured thus. At each casting it is observed whether the working takes the form intended. For A and B, a flame higher or lower than the hand's breadth set down by a quarter of a hand or more, or of any colour but the orange-yellow set down, is counted a deviation; for C, the object to be perceived is chosen each time by the assistant, by lot, from twelve small things and laid in the casket, the caster not knowing which, and a

perception that is wrong or that finds nothing is counted a deviation. Whether a casting deviates is decided by the recorder, by the standards set down and the slip of the lot. The number of deviations in a hundred castings is taken as the measure of steadiness.

A surrounding acts not only upon the drawing of power but upon what the working produces. A flame in a storm is bent by the wind, which is an action of the wind upon the fire and has nothing to do with the place whence the fire draws its power. To remove this the caster stands within the disturbance, but what he produces is formed within a small chamber, closed against the wind and graven on every wall with signs that bar the elements; and only the instant of forming is observed. The castings of comparison are formed in the same chamber, so that whatever the chamber itself may do is the same on both sides. Whether the chamber acts together with the disturbances was also tried. A second chamber was made, closed against the wind but without the signs that bar the elements, and some sets of A and B were cast with it in storm and within the array. In storm A deviated more in this chamber, but gave the same number of castings as in the graven one; within the array the two chambers gave the same. The signs, then, keep the disturbance from what is produced, and do not alter the drawing of power. In this paper the drawing of power is judged chiefly by the castings, the deviations serving as further witness.

The casting of one crystal from its first use until it goes dark I call a round; several rounds by the same caster, of the same working, in the same surrounding, I call a set. The numbers of castings given in the tables are the means of all the rounds in the place concerned, and the deviations are reckoned in each hundred castings. By the variation I mean the greatest difference between the means of the sets of comparison.

§ 2. Of the Place whence Power is Drawn

We cannot see whence a working draws its power; we see only in what disturbances it is disturbed. In this paper, therefore, the place whence power is drawn is judged by the disturbances alone. If two workings are disturbed or undisturbed alike by every disturbance, we have no means of telling their places apart; if a single disturbance disturbs the one and not the other, they differ at least in that part of their drawing which the disturbance reaches, and this I call drawing from different places. The more disturbances are tried, the more places can be told apart; and there are at least as many places as there are different manners of being disturbed among the workings tried. By "places" I mean such as the trials can distinguish; whether any of them may be divided further I do not consider. To judge in this way it is not necessary to assume any doctrine of Tiamat beforehand; a doctrine serves only to explain what the trials show.

§ 3. That Sameness of Place is Relative

By the rule of the preceding section it cannot be said absolutely whether two workings draw from one place; it depends upon the disturbances with which they are tried. If storm alone is tried, neither the law-working nor the string-spell is disturbed, and they cannot be told apart; if the tide alone, all three workings are disturbed, and again they cannot be told apart; only when the locking of lines is tried do the law-working and the string-spell part company. "The same place" and "a different place" are therefore spoken relatively to the disturbances tried. As more disturbances are tried, what was taken for one place may

be divided; but what has once been divided is never joined again, so long as the judgments of disturbance on which the division rests are not themselves overturned.

The friends of the doctrine of two layers say that law-workings and string-spells draw from one place because they have tried them only by storms and by the unbalancing of the elements, in which neither is disturbed, so that the two look alike. They did not see wrongly; they tried too few disturbances.

§ 4. How the Order of the Layers may be Read from the Manner of Disturbance

When several disturbances and several workings have been tried, let the disturbances that disturb each working be written in a row; this I call a table of disturbance.

If the layers lie one upon another, if each disturbance reaches one layer only, if the path by which each working draws is a continuous stretch beginning from the lowest layer, and if every working whose path passes a disturbed layer is found disturbed in the trials, then the table of disturbance must have a particular form. Let the workings be arranged according to the number of disturbances that disturb them: then the disturbances of each working include all those of the working before it. Such a table I call a nested table.

Conversely, if the table of disturbance is nested, an order of the layers agreeing with these conditions may be read from it thus. The disturbances of the working least disturbed are to be taken as reaching the lowest layer; those by which the next working exceeds it, as reaching the layer above; and so on. The layer at which each working takes hold is that reached by the uppermost of its disturbances. The form of the table shows only that the conditions can hold together; it does not by itself prove that the world is so made. And the order is fixed uniquely only if each working exceeds the one before by a single disturbance; where it exceeds it by two or more, the order of the layers these reach cannot be told from the table, and workings that take hold between them must be tried.

If the table is not nested — if, for example, each of two workings is disturbed by something that does not disturb the other — then one at least of the conditions fails: either the layers do not simply lie one upon another, or some disturbance reaches more than one layer, or the path of some working is broken; and the doctrine of layers must be amended. Here the doctrine may be overthrown.

§ 5. Of the Compounding of Losses

Since power passes layer by layer, each disturbed layer that it passes takes its own abatement. Let a crystal give n_0 castings in comparison, and n_i when the i-th layer alone is disturbed; I call $f_i = n_i/n_0$ the abatement of the i-th layer. Suppose that the power stored in a crystal is fixed, so that the number of castings is inversely as the expenditure of each, the remnant short of a casting not being counted; and suppose that the disturbance of each layer increases the expenditure of a casting in a fixed proportion, and not by a fixed amount, the proportions of the several layers being independent of one another. Then when several layers are disturbed together the number of castings a crystal gives should be

$$n = n_0 \times f_1 \times f_2 \times \dots \tag{1}$$

where only the disturbed layers on the working's path enter the product, the abatement of any layer not on the path being reckoned as one. This rule I call the compounding of losses. It is not among the three premisses of § 12 but is added to them, and Table III of § 9 is a test of it. Were the disturbance of each layer merely to add a fixed amount to

the expenditure, the castings under several disturbances would not be the product but somewhat more; and Table III decides between the two. By the additive reckoning the castings under several disturbances satisfy $1/n = 1/n_1 + 1/n_2 - 1/n_0$.

The compounding of losses says only how the abatements of the several layers combine for one and the same working; whether one layer abates different workings alike is another matter, and does not follow from it. In Table I the abatement by the locking of lines is 31/40, or about 0.78, for the elemental working, and 29/41, or about 0.71, for the law-working; both are of the second ring, but the difference lies at the edge of the variation and cannot be settled. The abatement by the tide, taken from the second tide year (§ 8), is 29/40 for the elemental working and 30/41 for the law-working, both about 0.73, but only 13/22, or about 0.59, for the string-spell, which is plainly different. The abatement of one layer, then, may differ from working to working. How it varies with the kind of working, its ring and its expenditure, the workings tried here, all of low rings, permit no finer division, and I leave this point to others.

PART II. — OF THE TRIALS

§ 6. The Three Workings

The three workings tried were all of low rings, lest the expenditure be too great and the casters, being wearied, disturb the counts.

A, the elemental working: a flame some hand's breadth high, gathered from the element of fire; of the second ring.

B, the law-working: a flame of the same height, made by plucking the lines of law; also of the second ring. A and B produce the same thing by different paths, and are therefore fit to be compared.

C, the string-spell: a perceiving string-spell of a low ring, cast by a fellow of the White Tower Society versed in the lower string-spells, by leave of the Tower.

Three casters were engaged for each of A and B; for C there was but one who could cast it. That C came from a single hand, and that only one string-spell was tried, are two defects of this paper.

§ 7. The Three Disturbances

First, the elemental disturbance: the stormy airs where the elements are out of balance. In the Months of Autumn Dusk and of Frostfall the element of air runs wild in the stormy airs north of White Tower, and the other elements fall short. We sailed into these airs and cast aboard the ship. Whether the elements were out of balance was judged by four crystal plates of the four elements hung from the mast: the trials began when the plate of air shone and the other three were dim, and when the four were of like brightness the ship sailed out and the castings served for comparison.

Second, the disturbance of laws: within a line-locking array (Fig. 1), in which several lines of law are locked. The lines locked were to have nothing to do with those used by the three workings; had the very line of fire been locked, A and B could not have been cast at all, which would show only that the line of fire was locked and nothing of the place whence power is drawn. That they had nothing to do with them was judged first by the customary division of the lines among law-mages and in the manuscripts of the Tower, by which the lines that serve a flame and a perception belong to other divisions than those

locked, and have never been found joined with them; and then by trial: after each group was locked, a candle lit without any working was set in the chamber, and the height and colour of its flame did not change, nor did the look of the things in the casket, so that the lines locked act directly neither upon the flame nor upon the thing perceived. After each opening of the array a law-mage first tried to pluck the locked lines, and only when they would not move were they held locked. The lines locked were changed seven times in all; see Table IV. That the locking reaches only the layer woven of the lines of law is one of the premisses of § 12; these trials support it, but do not prove it.

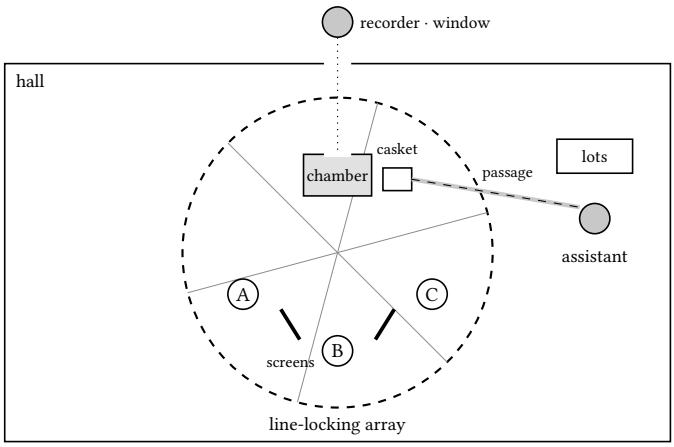


FIG. 1. The arrangement of the trial by locking. The array stands in the middle; the three casters stand within it some scores of paces apart, with screens between them; the flames are formed in the chamber barred against the elements within the array, and the recorder watches the chamber through a window from outside the hall; C's casket stands beside the chamber; the assistant, outside the array, draws a lot to decide whether it shall be opened, and changes the thing in the casket through a passage from outside.

Third, the tidal disturbance: the height of the tide of mana. The tide comes once in ten years, and its height lasts no more than some ten days. The days of the height were judged by the tide-reports of the cities and the common feeling of the casters; the charged crystals also glow faintly of themselves about the height, which is further evidence. Since the crystals glow of themselves, the point at which the light is reckoned out might be moved. We therefore kept the crystals that had gone dark at the height until the tide had ebbed, and looked again: their faces still showed no grain, and not a single casting could be had from them; and crystals that had gone dark in storm or within the array were the same when taken out. None of the three disturbances, then, alters the relation between the going out of the light and the power remaining: when the light is out, what remains is short of a single casting.

§ 8. Of Comparison and Repetition

To every disturbance belongs a set of comparison, cast by the same caster, with the same batch of crystals and the same working, outside the disturbance. For the trial by storm the comparison was cast in calm air after the same ship had sailed out of the stormy airs, the ship sailing in and out by turns on the same day, with some rounds cast each time; for

the trial by locking, in the same hall when the array was not opened; for the trial by tide, in the same place, once in the month before the tide and once after it had ebbed, the two agreeing, so that the height lay between them. The three sets of comparison agree with one another and are recorded together below.

The three workings were cast in the same place at the same instant: the three casters stood some scores of paces apart and cast at a signal, lest a difference of hours be mixed with the difference of workings. That casting at such distances causes no mutual disturbance was ascertained beforehand. The three casters of A and of B took their turns. Whether the array was opened was decided by the assistant by lot, the casters not knowing beforehand, lest expectation affect the casting. The procedure was the same whether the array was opened or not: the assistant went to the urn of lots outside the array, and the law-mage tried the lines, as always; when the array was not opened the lines moved under his hand, and he said nothing. The figures of the array were graven beneath the floor of the hall, out of the casters' sight. The recorder who judged the deviations watched the chamber through a window from outside the hall, and he too did not know whether the array was open; the lots were unsealed only when the day's trials were done. The chamber had a window on the side towards the outside of the hall; C's casket stood beside the chamber, and the assistant changed the thing in it through a passage from outside the array, never entering it. Screens stood between the three casters, and between rounds they rested apart and did not speak, so that none could see when another stopped because his crystal had gone dark.

The casters practised for a year before the trials proper began, until the number of castings a crystal gave them in comparison varied by no more than two from day to day. The castings and deviations of each round were recorded by two clerks separately, and entered only when the two agreed; a round on which they disagreed was discarded and cast again. In twenty years some forty rounds were so discarded, most of them in storm; when they are restored, each with the mean of what the two clerks recorded, no number in Table I changes by more than one, and the conclusions are unchanged.

The trials by storm and by locking were made both at White Tower and at Lataki, and the two places agree (Table V); White Tower tried about twice as many rounds as Lataki, so that the combined numbers of Table I mostly agree with those of White Tower. The trial by tide must await the tide years. In the first, C was tried for forty rounds only, and that as the height was passing; a crystal gave on the mean 19 castings, against 21 in comparison that year, only two fewer, which is within the variation of the comparisons, and nothing could be decided. In the second tide year eighty rounds were cast while the height was full (the nine days within four of the height, some nine rounds a day), giving on the mean 13 castings against 22 in comparison, and the difference was settled. The daily counts of C in the second tide year are shown in Fig. 2. A and B were tried in full during the height in both years: in the first, whose tide was the weaker, A gave 31 castings and B 32; in the second, A 29 and B 30. The column of the tide in Table I combines the two years; for C, the 19 castings of forty rounds and the 13 of eighty make 15.

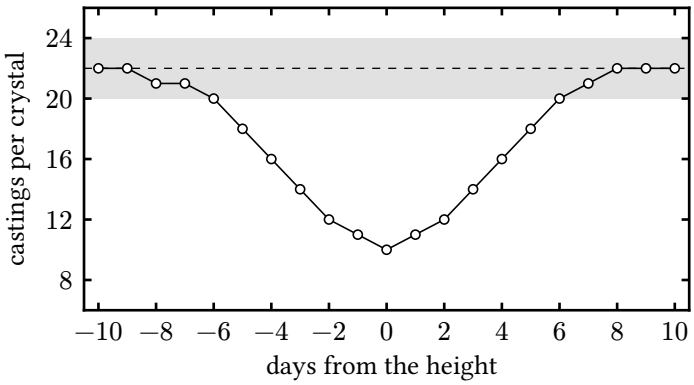


FIG. 2. Daily counts of the string-spell about the height of the second tide year. The abscissa is the number of days from the height; the ordinate, the number of castings a crystal gives. The broken line is the mean of comparison; the grey band, its variation.

§ 9. Results

Table I gives the expenditure and steadiness of the three workings in the three disturbances. In each place the first number is the number of castings a crystal gives, the second the number of deviations in a hundred castings; each place rests on some two hundred rounds, and that of C in the tide on a hundred and twenty. Among the sets of comparison the first varies by no more than two, the second by no more than three.

TABLE I. Expenditure and steadiness of the three workings under disturbance.

Working	*Comparison*	*Storm*	*Locking*	*Tide*
A, elemental	40 / 3	28 / 21	31 / 14	30 / 17
B, of law	41 / 2	40 / 3	29 / 18	31 / 15
C, string-spell	22 / 4	22 / 4	21 / 5	15 / 19

A difference from comparison greater than the variation among comparisons is reckoned a disturbance, which gives Table II.

TABLE II. Disturbed or not (● disturbed, ○ not disturbed).

Working	*Storm*	*Locking*	*Tide*
A, elemental	●	●	●
B, of law	○	●	●
C, string-spell	○	○	●

The sign ○ says only that no disturbance could be discerned within the power of these trials, not that there is none whatever. The three casters of A, and the three of B, agree each with their combined result. The three workings are disturbed in three different manners, so that by § 2 there are at least three places; and Table II is a nested table, so that by § 4 an order of the layers agreeing with the conditions may be read from it.

In the second tide year we cast the three workings also within the array, to see what follows when two disturbances are laid together; this gives Table III. All its numbers are of that year: comparison and locking alone were cast after the tide had ebbed, tide alone and both together during the height. The column “Both” is observed; the last, “By (1)”, is

the number of castings computed by (1) of § 5, the locking of C, which by Table II does not disturb it, being reckoned at an abatement of one.

TABLE III. Castings of a crystal in the second tide year when locking and tide are laid together.

Working	*Comparison*	*Locking alone*	*Tide alone*	*Both*	*By (1)*
A, elemental	40	31	29	22	22.5
B, of law	41	29	30	21	21.2
C, string-spell	22	21	13	13	13

Taking separately the seven groups of locked lines tried at White Tower, B was disturbed under every group and C under none (Table IV). Under the sixth and seventh groups, which locked three lines, B gave 27 and 28 castings, and under the groups of two from 28 to 30, as though the more lines were locked the more B were disturbed; but the number of the lines and which lines they were changed together. In the two pairs where one line only was added, the sixth group against the second and the seventh against the fourth, B gave but one casting fewer, which is within the variation. Whether more lines disturb more, these trials cannot settle. C was never disturbed.

TABLE IV. The seven groups of locked lines at White Tower, and the castings of B and C (locking / comparison).

Group	*Lines locked*	*B, of law*	*C, string-spell*
I	sound, weight	29 / 41	21 / 22
II	light, stillness	28 / 41	22 / 22
III	moisture, sinking	30 / 41	21 / 22
IV	shape, hardness	29 / 41	22 / 22
V	lightness, floating	30 / 41	21 / 22
VI	sound, light, stillness	27 / 41	21 / 22
VII	weight, shape, hardness	28 / 41	22 / 22

TABLE V. The trials by storm and by locking at the two places (castings of a crystal, disturbed / comparison).

Working	*White Tower, storm*	*Lataki, storm*	*White Tower, locking*	*Lataki, locking*
A, elemental	28 / 40	29 / 40	31 / 40	30 / 40
B, of law	40 / 41	39 / 41	29 / 41	30 / 41
C, string-spell	22 / 22	22 / 22	21 / 22	22 / 22

§ 10. Objections

If the differences of Table I did not arise from the places whence power is drawn, they might arise from other causes. I have thought of six, and remove them in turn.

First, the expectation of the caster. A caster who knew in which disturbance he stood might, without intending it, put forth more or less strength. Storm and tide cannot be hidden from him, but the array can; the trials by locking were therefore opened by lot, the casters not knowing. The most important place in Table II — that B is disturbed by locking and C is not — was obtained in trials in which the casters did not know. By not knowing I mean not knowing before the array was opened, nor as it was opened (§ 8);

whether the array, once open, made any change that the caster could feel in the course of casting, these trials cannot wholly exclude.

Second, a difference among the crystals. Crystals of one vein do not all hold the same charge. The crystals used for each working in each surrounding were assigned by lot from the same batch. To measure the time a crystal takes to go dark it must be spent, and a crystal so measured cannot be used again; from each batch, therefore, twenty were first drawn, and in a calm place a caster held one and the same small light upon each until it went dark, and the time was recorded. A batch in which these times differed too much was not used at all, and two batches were set aside in this way.

Third, the single caster of the string-spell. Since C came from one hand only, it might be thought that this fellow was himself peculiarly proof against locking. He was versed also in law-workings, and we asked him to cast B within the array. He was disturbed like the rest: a crystal gave him 30 castings against 41 in comparison, as with the other three casters of B. It is not, then, because he is himself proof against every locking that he was undisturbed by it. Whether it is so only when he casts this one string-spell, these trials cannot tell: one caster only cast the string-spell, and one string-spell only was tried (§ 6).

Fourth, an action of the disturbance upon what is produced. This was removed in § 1: the flames were formed in the chamber closed against wind and barred against the elements, whose own action was also tried; and the object perceived by C, changed each time by lot, was always set at the same distance, in the same casket.

Fifth, an action of the disturbance upon the crystals themselves. If storm, locking or tide made a crystal leak its charge, the castings would be fewer for that reason. We set unused crystals of the same batch, uncast, in each of the three disturbances and in the places of comparison for a day, and then measured, in a calm place and in the manner just described, the time each took to go dark: it was the same in all. At the height the tide is everywhere, and there is no calm place outside it; for the tide, therefore, half of a batch was set in the height and measured after the ebb, and compared with the other half measured before the tide, and again they agreed; and in an ordinary year the same days of rest did not change the time to go dark. A hindrance arising only while power is drawn could not be found by setting crystals aside; the trial of the point of darkness in § 7 makes good some part of this.

Sixth, the changes of twenty years. In twenty years casters grow old and instruments wear. But the comparison for each disturbance was always cast in the same year, at the same place, by the same caster, and never compared across years; the changes of many years fall alike upon both sides. Within a year, weariness and practice were evened out in the storm by sailing in and out by turns, and in the tide by the comparisons before and after, between which the height lay. The array could be opened by lot, but storm and tide must be waited for, and the comparisons for these two are never so clean as those for the array.

§ 11. That the Doctrine of Two Layers cannot account for the Table

By the doctrine of two layers the lines of law are Tiamat, and B and C both draw from them. Then the locking that disturbs B must disturb C; but C is not disturbed.

If it be said that C draws from another part of the lines, which the array happened not to lock, then locking other lines ought sometimes to disturb C. We changed the locked

lines seven times (Table IV): C was never disturbed, and B always was. If it be said that the tide disturbs C by way of the lines of law, then an array that locks the lines ought to stop, or at least to alter, the action of the tide upon C. In the second tide year we cast C within the array, and it was disturbed by the tide exactly as outside it (Table III).

Without further assumptions the doctrine of two layers cannot yield a result in which B is disturbed by locking and C is not; and the two assumptions just considered both disagree with the trials. The doctrine of two layers, in the form that what draws from one place is disturbed alike, therefore cannot stand, at least as regards string-spells. Seven groups of locked lines do not exhaust the lines of law; should it be further assumed that the array can never lock the part the string-spell uses, these trials cannot exclude it. But such an assumption is made for this one place in the table, whereas the doctrine of layers accounts for the whole table by one account, and foretold Table III besides.

§ 12. The Three Premisses

I take the following three as premisses, and inquire whether they account for Table II.

First, the source of mana lies lowest. All the power that any working draws comes in the end from the Sea of Mana, the sea upon which, by the fragments, Masha wove her web; and the tide of mana rises there.

Second, between the source and the caster there are several layers lying one upon another. A working that takes hold at a certain layer must draw its power through all the layers beneath it; this I call its path. The higher it takes hold, the more layers it passes.

Third, each disturbance reaches one layer only. A working whose path passes a disturbed layer shows a discernible change in expenditure and steadiness; one whose path does not, is not affected. The storm reaches the uppermost layer, where the elements are; the locking reaches the layer woven of the lines of law; the tide rises from the source and reaches the lowest layer, next the source.

The premisses do not prescribe beforehand at which layer each working takes hold; that is for the trials to determine. By the method of § 4: C is least disturbed, by the tide alone, so that the tide reaches the lowest layer and C takes hold there; B is disturbed besides by locking, so that the locking reaches the layer above, and B takes hold there; A is disturbed besides by storm, so that the storm reaches the uppermost layer, and A takes hold there. The manner of disturbance of every working is a continuous stretch from the lowest layer upward, none passing over a middle layer; and this is the test of the premisses by Table II. The three layers spoken of are the three that these trials can distinguish; whether any of them may be divided further I do not consider.

The compounding of losses of § 5 may be tested by Table III. In the second tide year A gives under locking alone 31/40 of its comparison, under tide alone 29/40, so that under both it should give 40 × 31/40 × 29/40, or about 22.5 castings; B should give about 21.2; C does not pass the layer of the lines, locking takes nothing from it, and under both it should give what it gives under tide alone, namely 13. The observed numbers in Table III are 22, 21 and 13, each within one casting of the product. Were each layer merely to add a fixed amount to the expenditure, A should give about 24 and B about 23, two more than observed, at the edge of the variation; Table III lies nearer the product. Since the compounding of losses was not used to fix the layers at which the workings take hold, this is a test apart.

§ 13. The Three Words of the Fragments

I return now to the three words of the fragments (Table VI). The first word the fragments use where they speak of what was woven upon the sea: it denotes the whole that Masha wove, which is Tiamat. The second they join with "silver lines" and "that may be plucked": it denotes the layer woven of the lines the law-mage plucks, which I call the Web of Laws. The third they join with words of straining and clearing: it seems to denote a layer between the sea and the web, which strains the wild mana clear. Former interpreters rendered all three as Tiamat, and so took the lines of law to be Tiamat. Compared with Table II, what the second word denotes is just the layer the locking reaches and B takes hold of; what the third denotes is just the layer the tide reaches and C takes hold of. In Table VI a passage repeated in several copies of the fragments is counted once only. That the three words denote three things is my interpretation, not a consequence of their number; they might be synonyms, or differ only as figures of speech. That the interpretation agrees with Table II is a further witness for the doctrine of layers.

TABLE VI. The uses of the three words in the fragments.

Word	*Uses*	*Words commonly joined with it*	*Denotes*
First	eleven	weave, sea, Masha	Tiamat entire
Second	seventeen	silver, line, pluck	the Web of Laws
Third	six	strain, clear, beneath	the lower part of Tiamat

The world, then, should be thus (Fig. 3). Lowest is the Sea of Mana; upon it Tiamat is laid, whose lower part strains the mana clear and divides its streams, and whose upper part is the Web of Laws; and above the web lies the layer of the elements. The three workings take hold, in order, at the layer of the elements, at the Web of Laws, and at the lower part of Tiamat. How the lower part of Tiamat strains and divides, this paper cannot say. What it has found is the order of the layers, not their make.

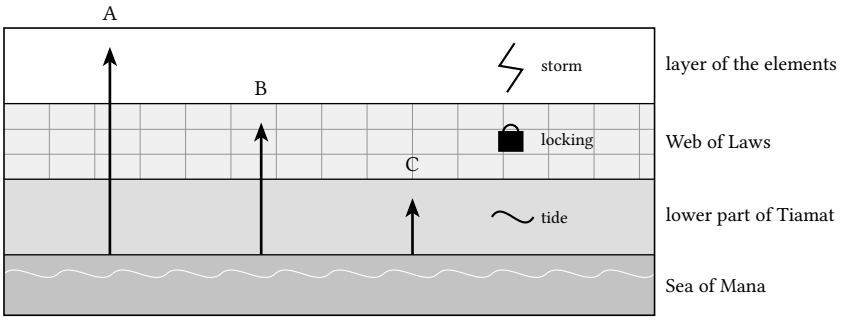


FIG. 3. The layers described by the doctrine of layers. Lowest is the Sea of Mana; above it the lower and upper parts of Tiamat, the upper being the Web of Laws; uppermost, the layer of the elements. The three upright lines are the paths of the three workings; the three marks show the layers reached by the three disturbances.

§ 14. Consequences

First. If every string-spell takes hold at the same layer as C, then by the three premisses no ward made by locking lines or by unbalancing the elements can keep a string-spell from being cast, of whatever ring. This is a prediction; only one string-spell of a low ring

has been tried. Sealed places and stores guarded by line-locking arrays alone, therefore, cannot be relied upon to keep string-spells from being cast; whether their walls and doors keep out the caster and what he would take is another matter.

Second. By the three premisses, the path of every working passes the lowest layer, and the tide of mana reaches every one. The three workings tried here all spent more and deviated more at the height of the tide; the casters' old saying, that in a tide year the hand is heavy, may have this reason. Whether other workings, when disturbed by the tide, likewise spend more and deviate more, I have not tried.

Third. Under the three premisses, and within what these trials can distinguish, the layer at which any working takes hold may be found by the three disturbances: one need only see which disturb it. Witchcraft, runes, holy words and the other kinds of working I have not tried, and I commend this method to those who come after. Should any working be disturbed by storm and not by locking, the table of disturbance would cease to be nested, and the doctrine of layers would have to be amended.

Conclusion

The doctrine of two layers was not without ground. As regards elemental workings and law-workings alone it is broadly right: both are disturbed by locking and by the tide, and no difference between them appears. Its error lies only in taking the Web of Laws for the whole of Tiamat.

The trials could not have been made but for the leave of the Tower, which allowed a fellow to cast string-spells, whose name by the custom of the Tower is not given; but for an old shipmaster, who twice a year for twenty years carried us into the storms; and but for my master Atherton. My master has never accepted the conclusion of this paper, yet every year he read my records with care and pointed out their faults; and the greater part of the comparisons here described were devised to answer his objections.